

NBA opening-spread relative value — model and strategy review

Supersedes nba_model_and_strategy_review.pdf, whose figures predate three corrections: the injury-report coverage frame (D186), the bet-time information leak (D199), and the trading frame (D207). The PDF is retained for history.

1. What this is

A market-anchored NBA sides strategy. A market-blind margin model prices every regular-season game; that model's **disagreement with the opening line** is then spent, at about a third of face value, as a correction to the opening line itself. Bets are one side of one game at −110, entered at the open, held to settlement. Flat 1 unit. No hedge, no exit, no calendar filter.

$$m_{\text{final}} = \text{open_margin} + f(x)$$

f is a ridge shrunk hard toward zero — toward *"the opener is right"* — fitted walk-forward on three features knowable at the open: the market-blind model's edge vs the opener, rest differential, and $|\text{open_margin}|$. Its edge coefficient is **0.33–0.37 in every fold**.

2. The frame, and why it is only seven seasons

Everything here is 2019-20 → 2025-26. Earlier seasons are excluded on principle, not on results.

The model has four inputs. Three go back to 2007-08. The fourth — the **daily NBA injury report**, which the availability leg is built from — **does not exist before 2018-12-17**, and covers only 63.7% of 2018-19. Availability composition is *half the production margin*, so before 2019-20 the model cannot run as designed: those seasons score a different, crippled system.

WHAT DATA ACTUALLY EXISTS — measured coverage of every model input, by season

The model has FOUR inputs. Three of them go back to 2007-08. The fourth — the daily injury report the availability leg is built on — does not exist at all before 2018-12-17. Eleven full seasons (2007-08 ... 2017-18) therefore score a CRIPPLED variant of the model, not the shipped one. Any figure in this repository spanning those seasons blends two different systems.



Pooling them was costing us in both directions. It diluted the result — the strategy reads **+9.11%** on the honest frame against **+7.57%** on the 14-season blend — and it made every interval a mixture of two different models.

The cost of that honesty is power. Seven seasons is not many, and it is why nothing below is statistically significant.

A second, tighter window matters for execution: a genuine **multi-book panel at the open exists for 2023-24 only** (7.74 books/game; 2024-25 and 2025-26 observe 1.00 and 1.03). Multi-book figures outside that season are modelled uplift.

3. Results

Priced at the opening spread, $k=9$ (max books observed), walk-forward selection.

	market-blind	market-offset
pooled ROI	+4.74%	+9.11%
cumulative	+49.0u	+80.9u
bets	1,035	888
season-clustered mean	+3.14%	+6.67%
95% CI (K=7)	[−5.79, +12.07]	[−2.75, +16.08]
profitable seasons	4/7	5/7

Neither is significant. Both intervals contain zero.

At a flat \$10,000 stake, the offset arm: **565 sessions, net \$809,335, \$1,432/day, Sharpe 1.1, win days 53%, max drawdown −\$240,301, edge 911 bps**. The blind arm: \$490,095, Sharpe 0.6, edge 474 bps.

Simulation performance — NBA opening-spread relative value (MARKET-OFFSET)

565 betting sessions across 7 seasons (2019-20 ... 2025-26) · 888 bets · net \$809,335 at a flat \$10,000 stake · edge 911 bps · tier k=9 raw (max books, no haircut)



The result is not evenly earned. The offset arm's first three seasons (2019-22) return **−\$30,637 at Sharpe −0.1**; the last four (2022-26) return **+\$839,972 at Sharpe 2.0** — and that recent block is also the block the architecture was developed on.

Why the summary is not cut to the last three or four seasons

The recent block looks much better: 2022-26 returns **+14.92%** against the frame's +9.11%, at Sharpe 2.0. That is a selection, not a regime.

window length	best window	mean of all windows of that length	selection buys
3 seasons	+16.95%	+9.03%	+7.92 pts
4 seasons	+14.92%	+9.33%	+5.59 pts
5 seasons	+13.47%	+9.94%	+3.53 pts
7 (the frame)	+9.11%	+9.11%	0.00

The average window returns ~+9% at every length. Shortening does not find a better regime, it finds a better draw — and the full frame is the only window whose selection cost is zero by construction. One five-season window even shows a significant interval; it is the best of three, which is the maximum of three draws rather than a finding. (D208)

On the measured-panel window (2023-26)

tier	market-blind	market-offset	delta
k=1 raw	+4.44%	+10.63%	+6.19
k=9 raw	+10.21%	+16.62%	+6.40
k=9 +haircut	+7.87%	+14.36%	+6.48

+6.4 ROI points at every tier including a single book — the gain is model, not shopping. And the quality improves more than the headline: positive **3/3** seasons against 2/3, best-season concentration **66%** against 84%, on **fewer** bets (460 vs 532).

4. Yes, we built our own forecasting model. It loses to the market.

This is the most useful negative result in the project, and the offset construction does not make sense without it.

Model, opening line and closing line on **identical games**, honest inputs:

source	log loss	vs the model
our market-blind model	0.59276	—
opening line	0.59228	we are 0.00048 worse
closing line	0.57870	we are 0.01406 worse

Beaten by the close decisively — 12.3% of its skill-above-a-coinflap — and by the opener narrowly, 0.5%. Expressed as the share of open-to-close information recovered: the blind model captures **−3.5%** (it moves the wrong way); the offset construction captures **+26.7%**.

So the market-blind model is not a discarded attempt. **It is the offset model's dominant input**, and the finding that it must be spent at a third of face value against a market anchor — rather than trusted on its own — *is* the strategy.

5. The bet-time information problem, and the fix

The availability leg was built on the **5PM injury report** and **pregame inactives**. Both publish *after* the opening line. Every open-priced figure predating D199 therefore gave the model information the bettor did not have.

Measured: **81.9%** of a team's out-set is already known from the prior report, but **18.1% is new on the day** (19.3% minutes-weighted). Rebuilding the out-set by carry-forward from the last report *strictly before* game day widened the model's gap to the market from **12.87% to 17.17%** — the late information was worth **33% of our entire deficit**.

The fix, gated and shipped. The composition leg now weights each player by **1 − P(out)**, forecast from as-of-open information only. Players last listed *Questionable* are out **28.9%** of the time; the old hard rule scored them 0.000 and was wrong in both directions. Gate: season-clustered delta -0.002265 , **CI [−0.0041, −0.0004] excludes zero**, $t = -3.39$, better **5/5 seasons**, calibration veto passed. **It recovers 52% of the leakage penalty** using only what is public at the open.

Critically, **the offset architecture survived this correction and the standalone model did not**: blind capture fell from $+0.075$ to **−0.104**; offset fell from $+0.313$ to **+0.267**. Because the ridge already spent the model edge at a third of face value, it was never leaning on the contaminated signal.

6. What would change our mind

- **2026-27, scored prospectively.** The architecture was developed on 2021-26; no retrospective procedure can make it out-of-sample. Both arms will run, on identical games, with the blind arm as the live control.
- **Two books captured at the open, from opening night.** The logger has never run in-season and odds_quotes is empty. If it is down on opening night the season's open-price record is unrecoverable.
- **Seven more seasons.** Not available. This is why the intervals are what they are.

7. Caveats

- Simulated on recorded prices. **No capital has ever been deployed.**
- Multi-book execution is counterfactual outside 2023-24.
- Sharpe is annualised on realised session count ($\sim 80/\text{season}$), never $\sqrt{252}$, which would inflate it 1.8×.
- The 2019-26 frame is too short to tune on: a best-of-five **random** game subset buys **+2.54 ROI points**.
- Neither arm is the production default. Promoting the offset construction is a re-certification and has not been done.
- The register (DECISIONS.md, 207 entries) records the rejections, the four team-name join bugs, the availability leak, two wrong frames, and a confidence interval that briefly claimed false significance. Most entries are failures.